dt 240 – 220 20

12 A

$GMm / r^2 = m\, r\, \omega^2$

$GM / r^3 = \omega^2$

$GM / r^3 = (2\pi / T)^2$

$T^2 = (4\pi^2 / GM)\, r^3$

$T^2 \propto R^3$

$T_X^2 / r_X^3 = T_Y^2 / r_Y^3$

$T^2 / r_X^3 = T_Y^2 / (4\, r_X)^3$

$T_Y = 8\, T$

So planet Y's orbital period is 8 times that of X. Planet Y completes only $\frac{1}{8}$ of an orbit for every 1 orbit of Planet X.

Hence in time T, planet Y only completes 0.125 of a revolution.